🔑 **Answer Key — Teacher Copy · Lesson 9-2**

MSTAR PRACTICE WORKSHEET

Lesson 9-2: Analyze Graphs of Relationships Between Two Variables

Grade 6 Mathematics · Standard 6.AT.11

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: D

Football tickets cost $45 each. What is the total cost of 3 tickets?

A $45

B $48

C $90

D $135

The total cost is 45 times the number of tickets, and 3 × 45 = 135.

Question 2 SELECTED RESPONSE — CORRECT: A

Which ordered pair belongs on the ticket graph for 2 tickets?

A (2, 90)

B (2, 45)

C (2, 47)

D (90, 2)

The number of tickets goes first, and 2 tickets cost 2 × 45 = 90 dollars, so the point is (2, 90).

Question 3 SELECTED RESPONSE — CORRECT: B

On the ticket graph, what should the two axes be labeled?

A. Number of tickets and price of one ticket

B. Number of tickets and total cost

C. Total cost and $45

D. Number of games and number of tickets

Each axis is labeled with one quantity from the problem — one axis shows the number of tickets and the other shows the total cost.

Question 4 SELECTED RESPONSE — CORRECT: D

The Sandia Peak tram travels 1,200 feet each minute. How far has it traveled after 3 minutes?

A. 1,200 feet

B. 1,203 feet

C. 2,400 feet

D. 3,600 feet

The tram covers 1,200 feet every minute, so after 3 minutes it has gone 3 × 1,200 = 3,600 feet.

Question 5 SELECTED RESPONSE — CORRECT: C

A table shows a train traveling 48 kilometers every hour: 1 hour, 48 km; 5 hours, 240 km. How many kilometers does it travel in 2 hours?

A. 48 km

B. 50 km

C. 96 km

D. 144 km

At 48 kilometers per hour, 2 hours gives 2 × 48 = 96 kilometers.

Question 6 SELECTED RESPONSE — CORRECT: D

As the number of tickets purchased increases, the total cost...

A decreases

B stays the same

C cannot be determined

D increases

Every extra ticket adds another $45, so the total cost rises as the number of tickets rises.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: B

A stadium shuttle carries fans from the parking lot. The table pairs the number of trips with the total fans carried: 2 trips with 96 fans, 5 trips with 240 fans, and 8 trips with 384 fans. On a graph with trips on the horizontal axis and fans on the vertical axis, which ordered pair belongs to 3 trips?

A. (3, 51)

B. (3, 144)

C. (3, 48)

D. (144, 3)

Every row of the table is 48 times the number of trips: 96 ÷ 2 = 48, 240 ÷ 5 = 48, and 384 ÷ 8 = 48. The independent variable, trips, is written first, and 3 × 48 = 144, so the ordered pair is (3, 144).

- **A:** 51 is not three times the rate. Every row runs 48 fans per trip, so three trips carry three times 48.
- **C:** 48 is the number of fans for ONE trip. Three trips carry three times that.
- **D:** The coordinates are swapped. Trips is the horizontal coordinate and fans the vertical.

Part B — correct answer: B

Which explanation justifies the ordered pair chosen in Part A?

A. The larger number always comes first in an ordered pair, so the fans are written before the trips.

B. The shuttle carries 48 fans on each trip, and trips come first in the ordered pair, so 3 trips pair with 3 × 48 = 144 fans.

C. The rate of 48 fans is what gets plotted, so 3 trips pair with 48 fans.

D. The trips and the fans are added, so 3 trips pair with 48 + 3 = 51 fans.

The ordered pair always lists the independent variable first, so trips come before fans. The relationship is a rate of 48 fans per trip, so the second coordinate is 3 × 48 = 144. Reversing the coordinates gives (144, 3), which reads as 144 trips carrying 3 fans.

Question 8 WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Reversed Coordinates

Ramona is graphing sales at the stadium pretzel stand. Pretzels cost $6 each, and her table shows 1 pretzel with $6, 2 pretzels with $12, and 3 pretzels with $18. She labels the horizontal axis 'number of pretzels' and the vertical axis 'total cost'. For 3 pretzels she plots the point (18, 3).

Explain Ramona's misconception and give the correct ordered pair for 3 pretzels.

Model answer: Ramona reversed the coordinates. Her horizontal axis is the number of pretzels and her vertical axis is the total cost, so the first coordinate has to be pretzels and the second has to be dollars. Three pretzels cost $3 \times 6 = 18$ dollars, so the point is (3, 18). The point she plotted, (18, 3), reads as 18 pretzels costing 3 dollars, which does not match any row of her table.

2 points	Student explains that the independent variable, the number of pretzels, is written first in an ordered pair and belongs on the horizontal axis, and gives the point (3, 18).
1 point	Student gives the point (3, 18) but does not explain that the independent variable is written first.
0 points	Student agrees with Ramona or gives an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.